Exploring Political, Demographic, and Socioeconomic Factors Behind Infrequent Voting

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Introduction

In the U.S. elections, the voter turnout remains to be a persistent challenge as millions of eligible do to participate. Although there have been efforts on increasing civic engagement, a significant portion of the population, particularly the infrequent voters, consistently fails to participate in voting. There is a need to understand the factors which influence voting behavior as it is crucial for political organizations to seek mobilization for underrepresented groups and to strengthen democratic participation. To investigate this issue, this case study analyzes survey data from FiveThirtyEight and Ipsos (2020) which includes responses from more than 5000 U.S. citizens on their voting habits, political affiliations, and demographic backgrounds. The data set contains variables such as age, race, education, income, political party affiliation (Democrat, Republican, Independent, etc.), and past voting behavior (categorized as "always", "sporadic," or "rarely/never" voters).

The goals of this study are: (1) to examine whether political party affiliation influences voting frequency, testing the hypothesis that partisans (Republicans and Democrats) are more consistent voters than Independents or those with no party preference; and (2) to develop a predictive model that identifies infrequent voters based on demographic and political characteristics, which can then be used to classify 500 new community members for targeted outreach efforts.

The impact of predictors such as age, education, income, race, and political affiliation on voting behavior which is the response variable will be examined by fitting statistical models—including logistic regression. In addition, interactions between variables are explores to further explain disparities in the voter turnout. This analysis aims to provide insights which are actionable for advocacy groups and political campaign which will hopefully help in designing more effective strategies to reduce participation gaps in future elections.

Data Description & EDA

The data set is from a survey from September 2020 which was conducted by FiveThirtyEight and Ipsos. It collected responses from 8,327 U.S. citizens through Ipsos's KnowledgePanel, and through a matching process withe the Aristotle's voter files using personal identifiers, 5,355 respondents (64% of the original sample) were successfully linked to verified voting records. To prevent under representation of infrequent voters who are less likely to appear in voter files, the researchers added

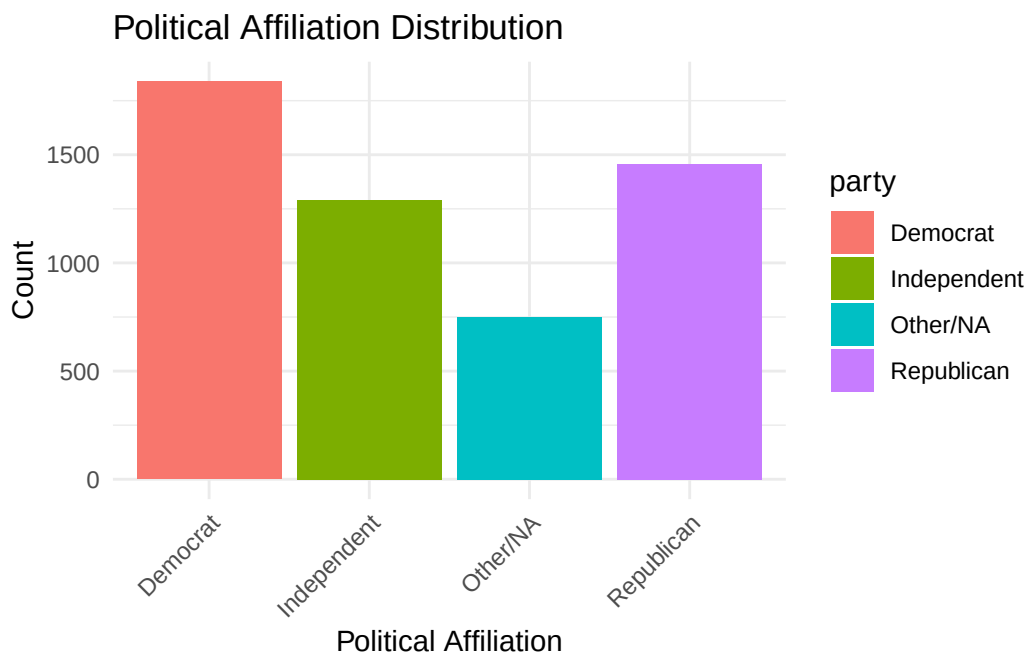
back 597 respondents who didn't match voter records but self-reported as rarely/never voters. This then created a preliminary sample of 5,952 cases (5,355 matched + 597 unmatched). After excluding respondents who were only eligible to vote in three or fewer elections, the data set contained 5,836 cases. Finally, removing observations with missing values in key variables resulted in the analytical sample of 5,336 complete cases used in this study. In R, we recoded the voter_category variable into an infrequent_voter indicator (1 for rarely/never voters, 0 otherwise), and combined missing values and categories 4 and 5 of the Q30 variable into a single category labeled 4 (Q30_combo_45na) to simplify analysis. In addition, we also created a new party variable by mapping values of Q30: coding 1 as "Republican", 2 as "Democrat", 3 as "Independent", and grouping 4, 5, and missing values as "Other/NA" to facilitate interpretation of political affiliation. This multi stage process balanced data quality with representation, particularly for low-propensity voters who might otherwise be excluded from analysis. Furthermore, we will explore variables included in the data set such as age, education, race, gender, household income, and political affiliation.

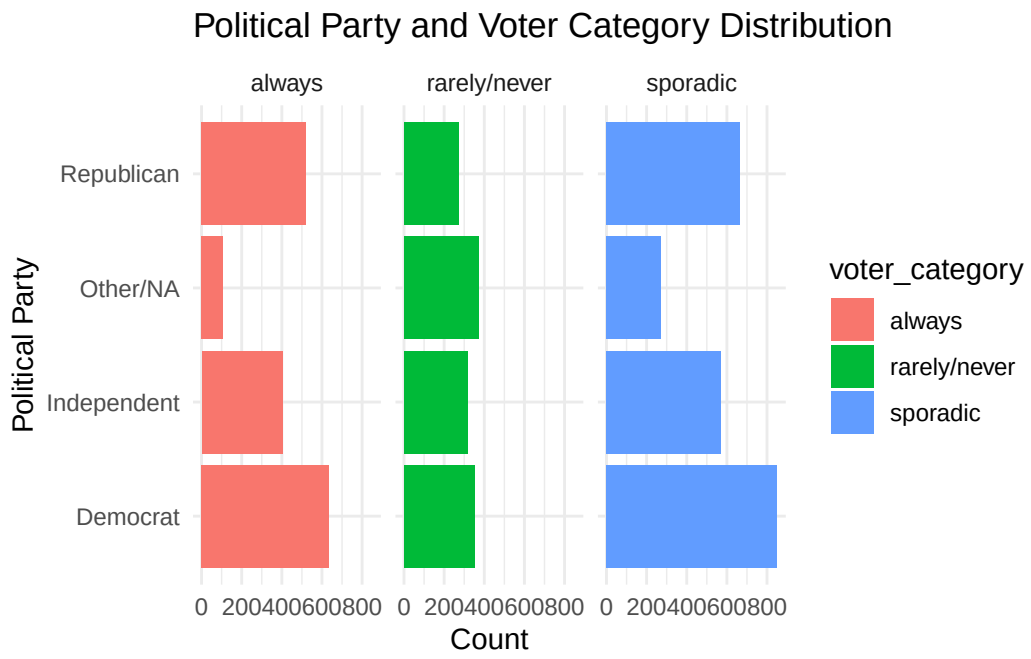
Table 1: Voting Frequency by Political Party Affiliation

party	always	rarely/never	sporadic	total	always_pct	sporadic_pct	rarely_pct
Democrat	635	352	851	1838	34.54842	46.30033	19.15125
Independent	403	316	572	1291	31.21611	44.30674	24.47715
Other/NA	109	370	270	749	14.55274	36.04806	49.39920
Republican	521	274	663	1458	35.73388	45.47325	18.79287

As we explore and examine the relationship between political party affiliation and voting frequency among U.S. citizens using survey data, the 'party' variable was recoded based on responses to Q30 (political party affiliation), as follows 1 to Republican, 2 to Democrat, 3 to Independent, and 4 to Other/NA (This includes missing, invalid, or non-response answers). The 'voter_category' variable was used to classify voters into three groups based on their voting frequency: (1) Always: Voters who vote in every election, (2) Sporadic: Voters who vote occasionally but not in every election, and (3) Rarely/Never: Voters who seldom or never vote.

Then, we calculated the total number of respondents for every party, and the percentage of respondents in each voting frequency category (Always, Sporadic, Rarely/Never) for each party. Two bar plots below were created to show the relationship of these variables. The first bar plot shows the total number of voters. The second bar plot shows the number of voters per political party then showed the frequency per voter category. These plots reflect the values in Table 1.





We found out that:

Republicans had a total of 1458 voters and had:

- 521 always voters (35.73%);
- 663 sporadic voters (45.47%); and
- 274 rarely/never voters (18.79%).

Democrats had a total of 1838 voters and had:

- 635 always voters (34.55% of Democrats);
- 851 sporadic voters (46.30%); and
- 352 rarely/never voters (19.15%).

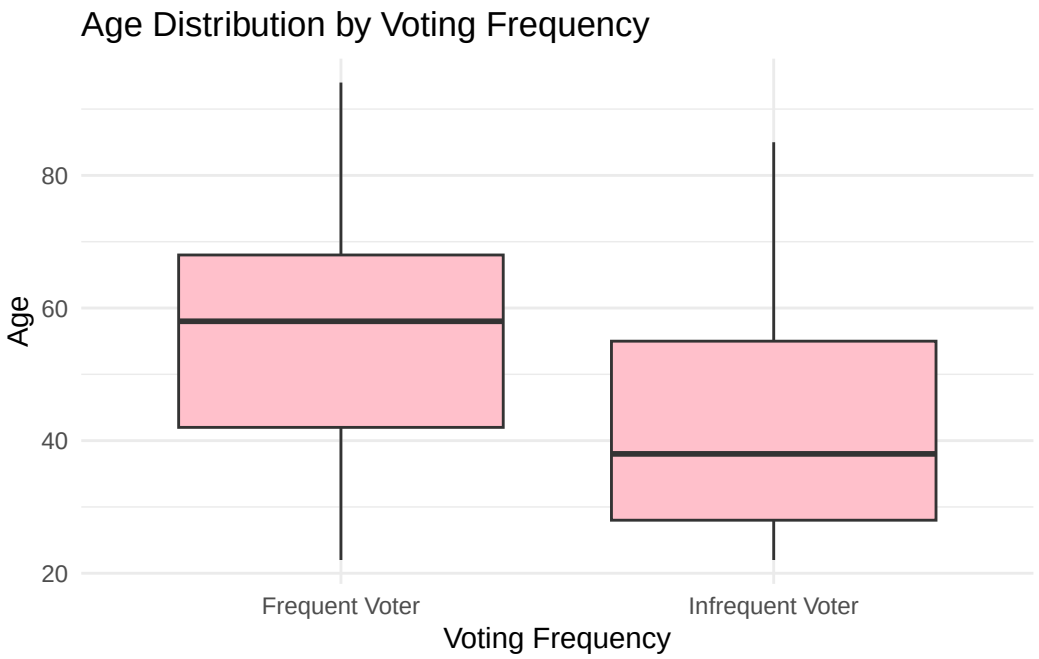
Independents had a total of 1291 voters and had:

- 403 always voters (31.22%);
- 572 sporadic voters (44.31%); and
- 316 rarely/never voters (24.48%).

Other/NA had a total of 749 voters and had:

- 109 always voters (14.55%);
- 270 sporadic voters (36.05%); and
- 370 rarely/never voters (49.40%).

Moreover, we explored one of the variables age.



We found out that infrequent voters (on the right) have lower median. It also has a narrower IQR when we compare it to frequent voters. Moreover, the lower and upper bounds of the boxplot are downward shifted meaning that infrequent voters skew younger. On the other hand, frequent voters (on the left) that they have a higher median. They also have a wider spread compared to the infrequent voters. With these, we can see that older individuals tend to vote in elections. However, this needs to be confirmed as we conduct the analysis.

Statistical Analysis and Results

Statistical Methods

The relationship between political party affiliation and voting behavior is being examined. The main explanatory variable is political party affiliation and is categorized into four groups: Republican, Democrat, Independent, and Other/NA. To ensure proper handling of incomplete responses, we treated values of -1 or "NA" as missing data. We created a response variable 'infrequent_voter' which is binary to indicate if a respondent rarely or never votes. A value of 1 represents a "rarely/never" voter and 0 represents all the other respondents. Through this, we allowed the focus to be on predicting who votes infrequently. Furthermore, categories 4, 5, and missing values in the political party variable were combined into one single category labeled as 4. This effectively simplifies the model by grouping low-frequency and missing responses.

We then generated the summary statistics. Furthermore, we checked to see whether any violations were present in the data (See R Appendix). To assess potential relationships among the variables, we tested the relationship between the combined political party variable (Q30_combo_45na) and 'voter_category' using a chi-square test.

A logistic regression model was then fitted to predict 'infrequent_voter' using 'Q30_combo_45na' (political party), 'ppage' (age), 'educ' (education), 'race', 'gender', and 'income_cat' (income category) as explanatory variables.

By conducting a backward stepwise selection we were able to systematically remove predictors which were non-significant based on Akaike Information Criterion (AIC). Through this, we achieved a more parsimonious or restraining model.

The log-odds estimates, standard errors, test statistics, p-values, and 95% confidence intervals were included in the final model results. Lastly, the finalized model was used to a new data set which generated predicted probabilities and which identifies likely infrequent voters.

Interpretation of the p-value and Chi-squared test

Table 2: Chi-Squared Test Results for Q30_combo_45na and Voter Category

statistic	p.value	parameter	method
325.7642	0	6	Pearson's Chi-squared test

The null hypothesis (H_0) states that there is no association between political party affiliation and voting frequency. The political party affiliation does not have any significant effect on the voting frequency.

On the other hand, the alternative hypothesis (H_A) states that there is an association between political party affiliation and voting frequency. The political party affiliation does have any significant effect on the voting frequency.

The p-value in the chi-squared test is much smaller than the typical significance alpha level of 0.05, so we reject the null hypothesis. This suggests that political party affiliation and voting frequency

are not independent, and there is a statistically significant relationship between the two variables.

People's party affiliation appears to be associated with how frequently they vote which means that different political party affiliations (e.g., Republicans, Democrats, Independents, etc.) tend to have different voting behaviors (e.g., always voting, sporadic voting, rarely or never voting).

This is an important finding because it suggests that political party affiliation may be a significant factor when considering which individuals are infrequent voters. Furthermore, we explore other variables such as age, education, race, gender, and income.

Model Description

Table 3: Logistic Regression Results: Predicting Infrequent Voting

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.470	0.194	2.431	0.015	0.091	0.850
Q30_combo_45na2	-0.113	0.102	-1.103	0.270	-0.313	0.088
Q30_combo_45na3	0.281	0.102	2.739	0.006	0.080	0.482
Q30_combo_45na4	0.962	0.111	8.676	0.000	0.745	1.180
ppage	-0.052	0.002	-22.265	0.000	-0.057	-0.048
educHigh school or less	1.016	0.097	10.466	0.000	0.826	1.206
educSome college	0.277	0.095	2.908	0.004	0.090	0.463
raceHispanic	0.088	0.129	0.683	0.494	-0.165	0.342
raceOther/Mixed	0.398	0.162	2.459	0.014	0.079	0.713
raceWhite	0.035	0.109	0.323	0.746	-0.176	0.249
genderMale	0.139	0.072	1.931	0.054	-0.002	0.281
income_cat$40-75k	0.060	0.112	0.538	0.590	-0.159	0.279
income_cat$75-125k	-0.062	0.107	-0.578	0.563	-0.271	0.148
income_catLess than $40k	0.583	0.113	5.146	0.000	0.362	0.806

Logistic regression model was used and applied to predict the probability of being infrequent voter (rarely/never voting). The model is:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \cdot \text{Democrat}_i + \beta_2 \cdot \text{Independent}_i + \beta_3 \cdot \text{Other}_i + \beta_4 \cdot \text{Age}_i + \beta_5 \cdot \text{Educ}_i + \beta_6 \cdot Race + \beta_7 \cdot \text{Gender}_i + \beta_8 \cdot \text{Income}_i$$

where π_i is the probability that respondent i is an infrequent voter. Furthermore:

β_1, β_2, β_3 (Democrat, Independent, Other: Political party affiliation). The beta coefficients represents the log-odds of being an infrequent voter for each group compared to the baseline group.

β_4 (Age) represents the effect of age on the likelihood of being an infrequent voter.

β_5 (Education) represents the effect of education on being an infrequent voter.

β_6 (Race) represents the effect of race on infrequent voting behavior.

β_7 (Gender) represents the effect of gender on infrequent voting behavior.

β_8 (Income) represents the effect of income on infrequent voting.

This model estimates the log-odds of being an infrequent voter as a linear function of political party affiliation, age, education, race, gender, and income.

Furthermore, we also evaluated various models by systematically removing different predictors and comparing their AIC values (see R Appendix). The full model had the lowest AIC, which was lower than models with certain predictors excluded. Moreover, models that removed 'ppage', 'gender', 'race', 'income_cat', 'educ' had higher AIC values which suggests that removing certain predictors leads to a deterioration in model fit. These predictors are important for explaining the outcome variable. Therefore, the full model, which includes all predictors, appears to be the best model.

Interpretation of the regression table

Party (Q30_combo_45na):

1. Intercept (0.470) - The baseline log-odds of being an infrequent voter for the reference category (Republican) are 0.470. The 95% confidence interval (0.091, 0.850) suggests that we are 95% confident that the true intercept lies between 0.091 and 0.850.
2. Q30_combo_45na2 (Democrat) (-0.113) - Compared to Republican (reference group), Democrat has 0.113 lower log-odds of being an infrequent voter. This difference is not significant since the p = 0.27. The 95% confidence interval includes 0 (from -0.313 to 0.088), we cannot confidently say that this predictor has a significant effect on the outcome at the 95% confidence level.
3. Q30_combo_45na3 (Independent) (0.281) - Independent party has 0.281 higher log-odds of being an infrequent voter compared to Republican. This is significant since the p = 0.006. The confidence interval (0.080, 0.482) does not contain 0, this result is statistically significant at the 95% level. This means the effect is likely real.
4. Q30_combo_45na4 (Other/NA) (0.962) - Other/NA party has much higher log-odds (0.962) of being an infrequent voter compared to Republican. This difference is highly significant since the $p < 0.001$. The confidence interval (0.745, 1.180) does not include 0 so this effect is also statistically significant. This suggests that this category has a strong positive association with infrequent voting.

Party affiliation matters. For those individuals who are grouped into Other/NA (category 4), where the likelihood of infrequent voting is much higher.

Age (Continuous Predictor:)

1. ppage (Age) (-0.052) - Each additional year of age reduces the log-odds of being an infrequent voter by 0.052. Older people tend to vote more frequently. This is highly significant since $p < 0.001$. The confidence interval (-0.057, -0.048) does not contain 0 which indicates that this relationship is statistically significant. This means age is a significant predictor of infrequent voting.

Education:

1. educHigh school or less (1.016) - People with High School or less education have much higher odds of infrequent voting compared to those with a higher education level. This is highly significant since $p < 0.001$. The confidence interval (0.826, 1.206) does not include 0, indicating statistical significance. This suggests that individuals with less education are more likely to vote infrequently.

2. educSome college (0.277) - People with Some college education have moderately higher odds of infrequent voting. This is significant since $p = 0.004$. The confidence interval (0.090, 0.463) does not contain 0, making this effect statistically significant.

People with less education (high school or some college) have a higher likelihood of being infrequent voters.

Race:

1. raceHispanic (0.088) - Hispanic race is not significant ($p = 0.494$), meaning it does not have a statistically significant effect on infrequent voting. The confidence interval includes 0, (-0.165, 0.342), so we cannot conclude that being Hispanic is a statistically significant predictor of infrequent voting.

2. raceOther/Mixed (0.398) - Being of Other/Mixed race is associated with significantly higher odds of being an infrequent voter compared to the reference group (White). This is significant since $p = 0.014$. The confidence interval (0.079, 0.713) does not contain 0, this effect is statistically significant.

3. raceWhite (0.035) - White race is not significant since $p = 0.746$. This means that it does not have a statistically significant effect on infrequent voting. The confidence interval includes 0, (-0.176, 0.249), so we cannot conclude that being White is a statistically significant predictor of infrequent voting.

Being of Other/Mixed race is associated with higher odds of infrequent voting compared to the reference race (White).

Gender:

1. genderMale (0.139): Males have somewhat higher odds of being an infrequent voter, though the result is borderline significant ($p = 0.054$). The confidence interval (-0.002, 0.281) includes 0, this effect is not statistically significant.

Income:

1. income_catLess than $40k (0.583) - People with an income of less than $40k have higher odds of infrequent voting compared to those with higher income. This is highly significant ($p < 0.001$). The confidence interval (0.362, 0.806) does not include 0, so this effect is statistically significant.

2. income_cat$40-75k, $75-125k (Not significant) - These income categories are not significant which means that income in these ranges does not significantly affect the likelihood of being an infrequent voter. The confidence intervals (-0.159, 0.279) and (-0.271, 0.148) respectively include 0, these predictors are not statistically significant.

Low income is a strong predictor of infrequent voting.

Classifying 500 new community members

Table 4: Summary of Predicted Classes

Predicted Class	Count
0	437
1	63

Using the fitted model, we predicted voting frequency for 500 new community members. We classified them as "infrequent voter" if their predicted probability exceeded 0.5, and "frequent voter" otherwise. Among the 500 new community members, 437 were predicted to be frequent voters, while 63 were predicted to be infrequent voters.

Discussion

The goals of this case study were to determine whether there is a relationship between political party and voting frequency and to build a predictive model which identifies infrequent voters in the community allowing the organization to send targeted political mailings to individuals who are not low propensity voters. In this analysis, logistic regression, variable selection strategies, and model comparison techniques were used to developed a parsimonious model which has included political party affiliation, age, education, race, gender, and income as explanatory variables.

We first tested the significance of the predictors. Our findings are as follows:

- Party affiliation: Independents and Other/NA were significantly more likely to be infrequent voters compared to Republicans, while Democrats did not differ significantly.
- Age: Each additional year of age was associated with a significant decrease in the odds of being an infrequent voter.
- Education: Individuals with high school or less and some college were significantly more likely to be infrequent voters compared to those with higher education levels.
- Race: Individuals identifying as Other/Mixed race were significantly more likely to be infrequent voters. On the other hand, White and Hispanic races were not significant predictors.
- Income: Individuals earning less than $40k had significantly higher odds of being infrequent voters. Middle income categories were not significant.
- Gender: Gender was not a significant predictor.

These findings suggest that infrequent voting is more likely among individuals who are younger, have lower educational attainment, lower income, identify as Independent or Other/NA party, and those of Other/Mixed race.

Then, the model which was built was applied to a new data set of 500 community members and these members were classified them as either infrequent or frequent voters. Among the 500 new community members, 437 were predicted to be frequent voters, while 63 were predicted to

be infrequent voters. Furthermore, the model identifies associations, not causal relationships—factors like education and income may be associated with infrequent voting but are unlikely to be direct causes. The results are being able to be generalized to populations which are similar to the community studied. Caution should be conducted when applying the model to communities with different demographic or political profiles. We should be able to explore having to add behavioral or attitudinal variables to improve predictions or testing interaction effects between predictors. The model provides actionable insights for targeting voter outreach, especially among Independents, individuals with lower education and income, younger adults, and those identifying as Other/Mixed race.

R Appendix

```
voter_data <- read.csv("voter_data.csv", na.strings = c(-1, "NA"))

# Creating infrequent_voter
voter_data <- voter_data |>
  mutate(infrequent_voter = ifelse(voter_category == "rarely/never", 1, 0))
```

```
apply(voter_data[, c("ppage", "educ", "race", "gender", "income_cat",
"Q30", "voter_category") ], 2, FUN=function(x)mean(is.na(x)) * 100)
```

```
       ppage           educ           race         gender     income_cat
   0.0000000      0.0000000      0.0000000      0.0000000      0.0000000
         Q30 voter_category
   0.7496252      0.0000000
```

```
voter_data$Q30_combo_45na <- voter_data$Q30
ids <- which(is.na(voter_data$Q30) | voter_data$Q30 %in% c(4, 5))
voter_data$Q30_combo_45na[ids] <- 4
voter_data$Q30_combo_45na <- factor(voter_data$Q30_combo_45na, levels = c(1,
 ↪  2, 3, 4))
```

```
table(voter_data$Q30_combo_45na)
```

```
   1    2    3    4
1458 1838 1291  749
```

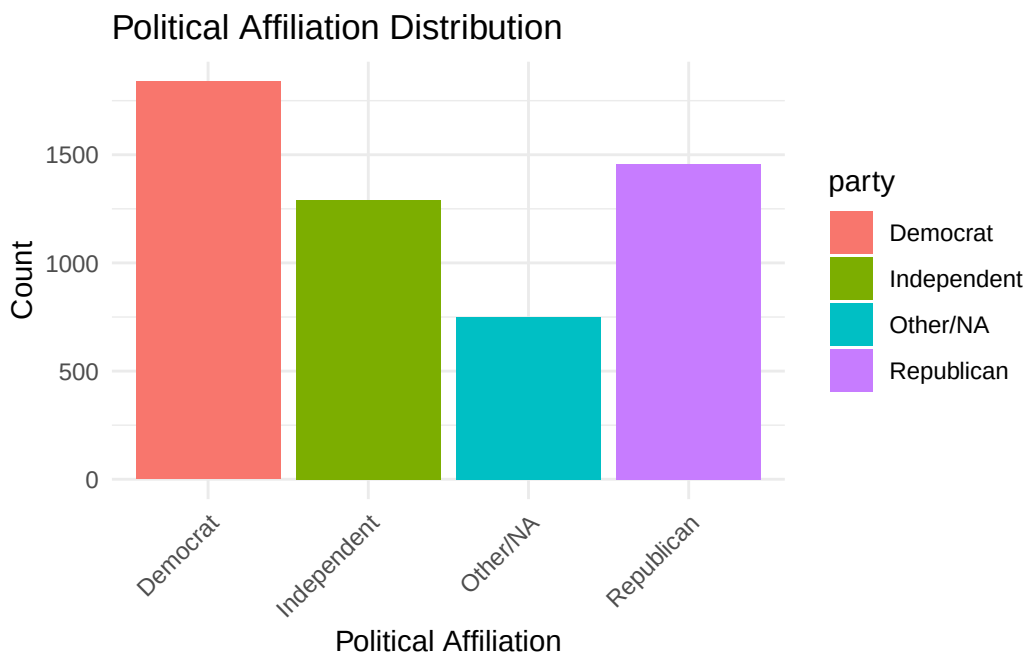
```
voter_data <- voter_data %>%
  mutate(
    party = case_when(
      Q30 == 1 ~ "Republican",
```

```r
      Q30 == 2 ~ "Democrat",
      Q30 == 3 ~ "Independent",
      Q30 %in% c(4, 5) | is.na(Q30) ~ "Other/NA"  # handling cases where Q30
 ↪  is 4, 5 or NA
    )
  )

party_voting_table <- voter_data %>%
  group_by(party, voter_category) %>%
  summarize(count = n(), .groups = 'drop') %>%
  pivot_wider(names_from = voter_category, values_from = count, values_fill
   ↪  = list(count = 0)) %>%
  mutate(
    total = `always` + `sporadic` + `rarely/never`,
    always_pct = `always` / total * 100,
    sporadic_pct = `sporadic` / total * 100,
    rarely_pct = `rarely/never` / total * 100
  )

kable(party_voting_table, caption = "Voting Frequency by Political Party
 ↪  Affiliation")
```

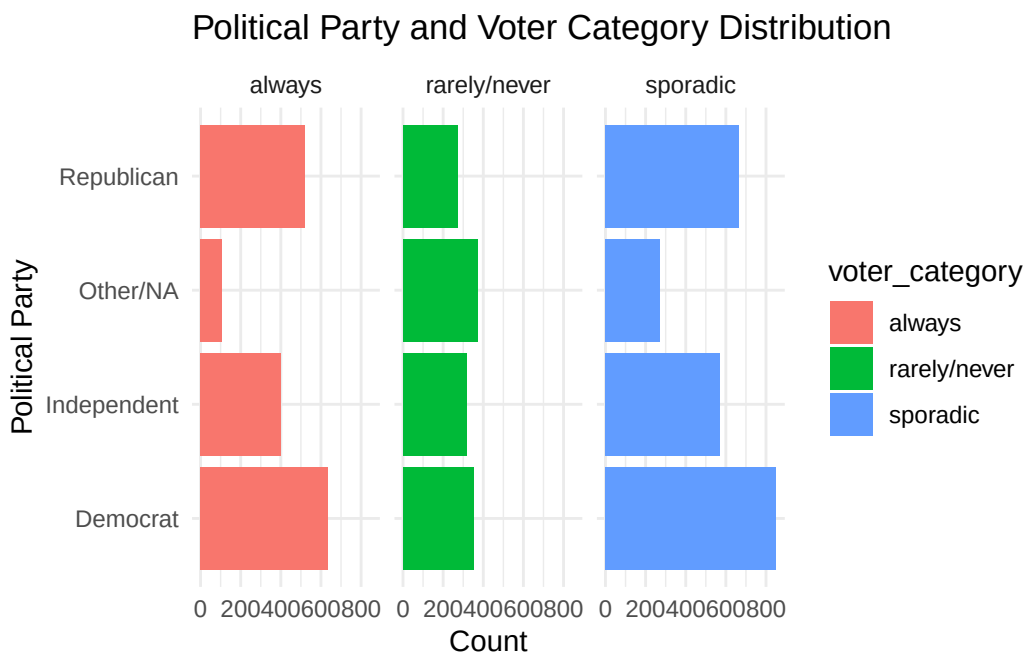
Table 5: Voting Frequency by Political Party Affiliation

party	always	rarely/never	sporadic	total	always_pct	sporadic_pct	rarely_pct
Democrat	635	352	851	1838	34.54842	46.30033	19.15125
Independent	403	316	572	1291	31.21611	44.30674	24.47715
Other/NA	109	370	270	749	14.55274	36.04806	49.39920
Republican	521	274	663	1458	35.73388	45.47325	18.79287

```r
ggplot(voter_data, aes(x = party, fill = party)) +
  geom_bar() +
  labs(title = "Political Affiliation Distribution", x = "Political
   ↪  Affiliation", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
ggplot(voter_data, aes(x = party, fill = voter_category)) +
  geom_bar(position = "stack") +
  facet_wrap(~ voter_category) +
  labs(title = "Political Party and Voter Category Distribution",
       x = "Political Party",
       y = "Count") +
  coord_flip() +
  theme_minimal()
```




```r
voter_data <- read.csv("voter_data.csv", na.strings = c(-1, "NA"))
voter_data <- voter_data %>%
  mutate(infrequent_voter = ifelse(voter_category == "rarely/never", 1, 0))

voter_data$Q30_combo_45na <- voter_data$Q30
ids <- which(is.na(voter_data$Q30) | voter_data$Q30 %in% c(4, 5))
voter_data$Q30_combo_45na[ids] <- 4
voter_data$Q30_combo_45na <- factor(voter_data$Q30_combo_45na,
                                    levels = c(1, 2, 3, 4))
```

```r
summary(voter_data)
```

```
    RespId           weight             Q1          Q2_1            Q2_2
Min.   :470001   Min.   :0.2298   Min.   :1   Min.   :1.000   Min.   :1.000
1st Qu.:472058   1st Qu.:0.7879   1st Qu.:1   1st Qu.:1.000   1st Qu.:1.000
Median :474158   Median :0.9673   Median :1   Median :1.000   Median :2.000
Mean   :474648   Mean   :0.9869   Mean   :1   Mean   :1.257   Mean   :1.722
3rd Qu.:476195   3rd Qu.:1.1668   3rd Qu.:1   3rd Qu.:1.000   3rd Qu.:2.000
Max.   :488325   Max.   :3.0386   Max.   :1   Max.   :4.000   Max.   :4.000
                                             NA's   :32      NA's   :31
     Q2_3            Q2_4           Q2_5            Q2_6            Q2_7
Min.   :1.000   Min.   :1.00   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:1.00   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
Median :2.000   Median :2.00   Median :1.000   Median :1.000   Median :1.000
Mean   :1.651   Mean   :2.19   Mean   :1.289   Mean   :1.817   Mean   :1.509
3rd Qu.:2.000   3rd Qu.:3.00   3rd Qu.:1.000   3rd Qu.:2.000   3rd Qu.:2.000
```

```
Max.   :4.000   Max.   :4.00   Max.   :4.000   Max.   :4.000   Max.   :4.000
NA's   :32      NA's   :30     NA's   :24      NA's   :27      NA's   :37
     Q2_8            Q2_9           Q2_10            Q3_1
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
Median :1.000   Median :2.000   Median :2.000   Median :2.000
Mean   :1.475   Mean   :2.111   Mean   :2.035   Mean   :1.923
3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:3.000   3rd Qu.:3.000
Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
NA's   :23      NA's   :33      NA's   :33      NA's   :32
     Q3_2            Q3_3            Q3_4            Q3_5
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000
Median :2.000   Median :3.000   Median :2.000   Median :2.000
Mean   :2.416   Mean   :2.668   Mean   :1.884   Mean   :2.041
3rd Qu.:3.000   3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:3.000
Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
NA's   :47      NA's   :48      NA's   :26      NA's   :40
     Q3_6            Q4_1            Q4_2            Q4_3
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
Median :2.000   Median :2.000   Median :2.000   Median :2.000
Mean   :2.082   Mean   :1.815   Mean   :1.847   Mean   :2.017
3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:3.000
Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
NA's   :30      NA's   :34      NA's   :35      NA's   :29
     Q4_4            Q4_5            Q4_6             Q5
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000
Median :2.000   Median :2.000   Median :2.000   Median :1.000
Mean   :2.454   Mean   :2.363   Mean   :2.046   Mean   :1.171
3rd Qu.:3.000   3rd Qu.:3.000   3rd Qu.:3.000   3rd Qu.:1.000
Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :2.000
NA's   :33      NA's   :39      NA's   :29      NA's   :29
      Q6              Q7             Q8_1            Q8_2
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:2.000   1st Qu.:1.000   1st Qu.:2.000   1st Qu.:2.000
Median :3.000   Median :1.000   Median :3.000   Median :3.000
Mean   :2.875   Mean   :1.192   Mean   :2.776   Mean   :2.778
3rd Qu.:3.000   3rd Qu.:1.000   3rd Qu.:4.000   3rd Qu.:3.000
Max.   :4.000   Max.   :2.000   Max.   :4.000   Max.   :4.000
NA's   :40      NA's   :55      NA's   :37      NA's   :40
     Q8_3            Q8_4            Q8_5            Q8_6
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:2.000   1st Qu.:1.000   1st Qu.:2.000   1st Qu.:2.000
Median :2.000   Median :2.000   Median :2.000   Median :2.000
Mean   :2.145   Mean   :2.021   Mean   :2.485   Mean   :2.237
3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:3.000
```

```
 Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
 NA's   :37      NA's   :30      NA's   :42      NA's   :20
      Q8_7            Q8_8            Q8_9            Q9_1
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
 Median :3.000   Median :2.000   Median :2.000   Median :1.000
 Mean   :2.811   Mean   :1.995   Mean   :1.833   Mean   :1.548
 3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000
 Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
 NA's   :38      NA's   :22      NA's   :33      NA's   :66
      Q9_2            Q9_3            Q9_4           Q10_1
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:3.000   1st Qu.:3.000   1st Qu.:2.000
 Median :2.000   Median :4.000   Median :4.000   Median :2.000
 Mean   :2.587   Mean   :3.249   Mean   :3.497   Mean   :1.922
 3rd Qu.:3.000   3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:2.000
 Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :2.000
 NA's   :72      NA's   :68      NA's   :75      NA's   :19
     Q10_2           Q10_3           Q10_4           Q11_1
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000
 Median :2.000   Median :2.000   Median :2.000   Median :2.000
 Mean   :1.795   Mean   :1.831   Mean   :1.994   Mean   :1.868
 3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000
 Max.   :2.000   Max.   :2.000   Max.   :2.000   Max.   :2.000
 NA's   :24      NA's   :25      NA's   :16      NA's   :19
     Q11_2           Q11_3           Q11_4           Q11_5
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:1.000   1st Qu.:2.000   1st Qu.:2.000
 Median :2.000   Median :2.000   Median :2.000   Median :2.000
 Mean   :1.979   Mean   :1.602   Mean   :1.864   Mean   :1.778
 3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000
 Max.   :2.000   Max.   :2.000   Max.   :2.000   Max.   :2.000
 NA's   :22      NA's   :18      NA's   :13      NA's   :13
     Q11_6            Q14             Q15             Q16
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
 Median :2.000   Median :2.000   Median :2.000   Median :1.000
 Mean   :1.981   Mean   :2.786   Mean   :2.317   Mean   :1.652
 3rd Qu.:2.000   3rd Qu.:5.000   3rd Qu.:3.000   3rd Qu.:2.000
 Max.   :2.000   Max.   :5.000   Max.   :5.000   Max.   :4.000
 NA's   :20      NA's   :189     NA's   :178     NA's   :53
     Q17_1           Q17_2           Q17_3           Q17_4           Q18_1
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.00
 1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:2.000   1st Qu.:2.00
 Median :2.000   Median :2.000   Median :2.000   Median :3.000   Median :2.00
 Mean   :1.665   Mean   :1.704   Mean   :2.348   Mean   :2.747   Mean   :1.97
 3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:4.000   3rd Qu.:2.00
```

```
Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :2.00
NA's   :45      NA's   :51      NA's   :47      NA's   :50      NA's   :41
    Q18_2           Q18_3           Q18_4           Q18_5           Q18_6
Min.   :1.00   Min.   :1.000   Min.   :1.00   Min.   :1.000   Min.   :1.000
1st Qu.:2.00   1st Qu.:2.000   1st Qu.:2.00   1st Qu.:2.000   1st Qu.:2.000
Median :2.00   Median :2.000   Median :2.00   Median :2.000   Median :2.000
Mean   :1.95   Mean   :1.943   Mean   :1.96   Mean   :1.985   Mean   :1.966
3rd Qu.:2.00   3rd Qu.:2.000   3rd Qu.:2.00   3rd Qu.:2.000   3rd Qu.:2.000
Max.   :2.00   Max.   :2.000   Max.   :2.00   Max.   :2.000   Max.   :2.000
NA's   :45     NA's   :42      NA's   :49     NA's   :49      NA's   :45
    Q18_7           Q18_8           Q18_9           Q18_10          Q19_1
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1
1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1
Median :2.000   Median :2.000   Median :2.000   Median :2.000   Median :1
Mean   :1.904   Mean   :1.808   Mean   :1.953   Mean   :1.943   Mean   :1
3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:1
Max.   :2.000   Max.   :2.000   Max.   :2.000   Max.   :2.000   Max.   :1
NA's   :49      NA's   :46      NA's   :42      NA's   :44      NA's   :3778
    Q19_2           Q19_3           Q19_4           Q19_5           Q19_6
Min.   :1       Min.   :1       Min.   :1       Min.   :1       Min.   :1
1st Qu.:1       1st Qu.:1       1st Qu.:1       1st Qu.:1       1st Qu.:1
Median :1       Median :1       Median :1       Median :1       Median :1
Mean   :1       Mean   :1       Mean   :1       Mean   :1       Mean   :1
3rd Qu.:1       3rd Qu.:1       3rd Qu.:1       3rd Qu.:1       3rd Qu.:1
Max.   :1       Max.   :1       Max.   :1       Max.   :1       Max.   :1
NA's   :2547    NA's   :2554    NA's   :3286    NA's   :3183    NA's   :2461
    Q19_7           Q19_8           Q19_9           Q19_10          Q20
Min.   :1       Min.   :1       Min.   :1       Min.   :1       Min.   :1.000
1st Qu.:1       1st Qu.:1       1st Qu.:1       1st Qu.:1       1st Qu.:1.000
Median :1       Median :1       Median :1       Median :1       Median :1.000
Mean   :1       Mean   :1       Mean   :1       Mean   :1       Mean   :1.082
3rd Qu.:1       3rd Qu.:1       3rd Qu.:1       3rd Qu.:1       3rd Qu.:1.000
Max.   :1       Max.   :1       Max.   :1       Max.   :1       Max.   :2.000
NA's   :3344    NA's   :3802    NA's   :4037    NA's   :5061    NA's   :18
     Q21             Q22             Q23             Q24             Q25
Min.   :1.000   Min.   :1.000   Min.   :1.00   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:2.000   1st Qu.:1.00   1st Qu.:1.000   1st Qu.:1.000
Median :1.000   Median :4.000   Median :2.00   Median :2.000   Median :2.000
Mean   :1.219   Mean   :4.317   Mean   :1.85   Mean   :2.001   Mean   :1.853
3rd Qu.:1.000   3rd Qu.:6.000   3rd Qu.:2.00   3rd Qu.:3.000   3rd Qu.:2.000
Max.   :3.000   Max.   :7.000   Max.   :3.00   Max.   :4.000   Max.   :4.000
NA's   :19      NA's   :4910    NA's   :97     NA's   :62      NA's   :22
     Q26             Q27_1           Q27_2           Q27_3
Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
Median :1.000   Median :1.000   Median :1.000   Median :1.000
Mean   :1.461   Mean   :1.274   Mean   :1.174   Mean   :1.344
3rd Qu.:1.000   3rd Qu.:2.000   3rd Qu.:1.000   3rd Qu.:2.000
```

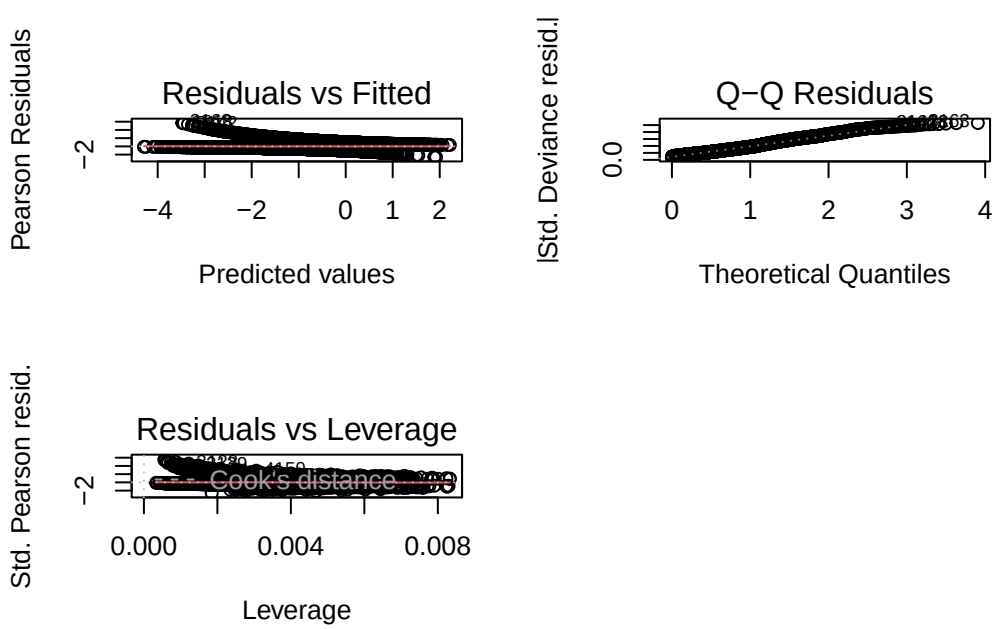
```
Max.   :4.000   Max.   :2.000   Max.   :2.000   Max.   :2.000
NA's   :21      NA's   :58      NA's   :48      NA's   :70
    Q27_4           Q27_5           Q27_6           Q28_1           Q28_2
Min.   :1.000   Min.   :1.000   Min.   :1.00   Min.   :1      Min.   :1
1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.00   1st Qu.:1      1st Qu.:1
Median :1.000   Median :1.000   Median :1.00   Median :1      Median :1
Mean   :1.233   Mean   :1.391   Mean   :1.29   Mean   :1      Mean   :1
3rd Qu.:1.000   3rd Qu.:2.000   3rd Qu.:2.00   3rd Qu.:1      3rd Qu.:1
Max.   :2.000   Max.   :2.000   Max.   :2.00   Max.   :1      Max.   :1
NA's   :45      NA's   :66      NA's   :46     NA's   :1619   NA's   :3558
    Q28_3           Q28_4           Q28_5           Q28_6           Q28_7
Min.   :1      Min.   :1      Min.   :1      Min.   :1      Min.   :1
1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1
Median :1      Median :1      Median :1      Median :1      Median :1
Mean   :1      Mean   :1      Mean   :1      Mean   :1      Mean   :1
3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1
Max.   :1      Max.   :1      Max.   :1      Max.   :1      Max.   :1
NA's   :3543   NA's   :3648   NA's   :3595   NA's   :3960   NA's   :3547
    Q28_8           Q29_1           Q29_2           Q29_3           Q29_4
Min.   :1      Min.   :1      Min.   :1      Min.   :1      Min.   :1
1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1
Median :1      Median :1      Median :1      Median :1      Median :1
Mean   :1      Mean   :1      Mean   :1      Mean   :1      Mean   :1
3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1
Max.   :1      Max.   :1      Max.   :1      Max.   :1      Max.   :1
NA's   :5225   NA's   :5010   NA's   :5188   NA's   :4969   NA's   :5088
    Q29_5           Q29_6           Q29_7           Q29_8           Q29_9
Min.   :1      Min.   :1      Min.   :1      Min.   :1      Min.   :1
1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1      1st Qu.:1
Median :1      Median :1      Median :1      Median :1      Median :1
Mean   :1      Mean   :1      Mean   :1      Mean   :1      Mean   :1
3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1      3rd Qu.:1
Max.   :1      Max.   :1      Max.   :1      Max.   :1      Max.   :1
NA's   :5123   NA's   :5292   NA's   :5191   NA's   :5142   NA's   :5230
    Q29_10          Q30             Q31             Q32             Q33
Min.   :1      Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
1st Qu.:1      1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
Median :1      Median :2.000   Median :1.000   Median :1.000   Median :2.000
Mean   :1      Mean   :2.354   Mean   :1.379   Mean   :1.376   Mean   :1.535
3rd Qu.:1      3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000
Max.   :1      Max.   :5.000   Max.   :2.000   Max.   :2.000   Max.   :2.000
NA's   :5151   NA's   :40      NA's   :3886    NA's   :3503    NA's   :3549
    ppage           educ               race              gender
Min.   :22.00   Length:5336        Length:5336        Length:5336
1st Qu.:37.00   Class :character   Class :character   Class :character
Median :54.00   Mode  :character   Mode  :character   Mode  :character
Mean   :51.77
3rd Qu.:66.00
```

```
Max.   :94.00

 income_cat         voter_category     infrequent_voter Q30_combo_45na
Length:5336         Length:5336        Min.   :0.0000   1:1458
Class :character    Class :character   1st Qu.:0.0000   2:1838
Mode  :character    Mode  :character   Median :0.0000   3:1291
                                       Mean   :0.2459   4: 749
                                       3rd Qu.:0.0000
                                       Max.   :1.0000
```

```r
table(voter_data$Q30_combo_45na, voter_data$voter_category)
```

```
  always rarely/never sporadic
1    521          274      663
2    635          352      851
3    403          316      572
4    109          370      270
```

```r
par(mfrow = c(2, 2))

plot(final_model, which = 1)  #Residuals vs Fitted
plot(final_model, which = 2)  #Normal Q-Q
plot(final_model, which = 5)  #Residuals vs Leverage

par(mfrow = c(1, 1))
```



```r
chisq_test <- chisq.test(table(voter_data$Q30_combo_45na,
 ↪  voter_data$voter_category))
chisq_test
```

```
    Pearson's Chi-squared test

data:  table(voter_data$Q30_combo_45na, voter_data$voter_category)
X-squared = 325.76, df = 6, p-value < 2.2e-16
```

```r
chisq_test <- chisq.test(table(voter_data$Q30_combo_45na,
 ↪  voter_data$voter_category))
chisq_table <- tidy(chisq_test)
knitr::kable(chisq_table, caption = "Chi-Squared Test Results for
 ↪  Q30_combo_45na and Voter Category")
```

Table 6: Chi-Squared Test Results for Q30_combo_45na and Voter Category

statistic	p.value	parameter	method
325.7642	0	6	Pearson's Chi-squared test

```r
full_model <- glm(infrequent_voter ~ Q30_combo_45na + ppage + educ + race +
 ↪  gender + income_cat, data = voter_data, family = "binomial")
```

```r
reduced_model <- stepAIC(full_model, direction = "backward", trace = FALSE)
```

```r
regression_table <- tidy(reduced_model, conf.int = TRUE)

knitr::kable(regression_table, digits = 3, caption = "Logistic Regression
 ↪  Results: Predicting Infrequent Voting")
```

Table 7: Logistic Regression Results: Predicting Infrequent Voting

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.470	0.194	2.431	0.015	0.091	0.850
Q30_combo_45na2	-0.113	0.102	-1.103	0.270	-0.313	0.088
Q30_combo_45na3	0.281	0.102	2.739	0.006	0.080	0.482
Q30_combo_45na4	0.962	0.111	8.676	0.000	0.745	1.180
ppage	-0.052	0.002	-22.265	0.000	-0.057	-0.048
educHigh school or less	1.016	0.097	10.466	0.000	0.826	1.206
educSome college	0.277	0.095	2.908	0.004	0.090	0.463
raceHispanic	0.088	0.129	0.683	0.494	-0.165	0.342

term	estimate	std.error	statistic	p.value	conf.low	conf.high
raceOther/Mixed	0.398	0.162	2.459	0.014	0.079	0.713
raceWhite	0.035	0.109	0.323	0.746	-0.176	0.249
genderMale	0.139	0.072	1.931	0.054	-0.002	0.281
income_cat$40-75k	0.060	0.112	0.538	0.590	-0.159	0.279
income_cat$75-125k	-0.062	0.107	-0.578	0.563	-0.271	0.148
income_catLess than $40k	0.583	0.113	5.146	0.000	0.362	0.806

```r
final_model <- glm(infrequent_voter ~ ppage + educ + race + gender +
 ↪  income_cat + Q30_combo_45na, data = voter_data, family = "binomial")
```

```r
final_model <- glm(infrequent_voter ~ Q30_combo_45na + ppage + educ + race +
 ↪  gender + income_cat,
                   data = voter_data, family = "binomial")

vif(final_model)
```

```
                   GVIF Df GVIF^(1/(2*Df))
Q30_combo_45na 1.197730  3        1.030528
ppage          1.106006  1        1.051668
educ           1.366059  2        1.081104
race           1.185434  3        1.028757
gender         1.033264  1        1.016496
income_cat     1.280521  3        1.042072
```

```r
new_data <- read.csv("new_community_members.csv")

new_data$Q30_combo_45na <- new_data$Q30
ids_new <- which(is.na(new_data$Q30) | new_data$Q30 %in% c(4, 5))
new_data$Q30_combo_45na[ids_new] <- 4
new_data$Q30_combo_45na <- factor(new_data$Q30_combo_45na, levels = c(1, 2,
 ↪  3, 4))

new_data_predictions <- predict(final_model, new_data, type = "response")

new_data$class <- ifelse(new_data_predictions > 0.5, "infrequent voter",
 ↪  "frequent voter")
```

```r
voter_data <- voter_data %>%
  mutate(party = case_when(
    Q30_combo_45na == 1 ~ "Republican",
    Q30_combo_45na == 2 ~ "Democrat",
```

```r
    Q30_combo_45na == 3 ~ "Independent",
    Q30_combo_45na == 4 ~ "Other/NA",
    TRUE ~ NA_character_
  ))

new_data <- new_data %>%
  mutate(party = case_when(
    Q30_combo_45na == 1 ~ "Republican",
    Q30_combo_45na == 2 ~ "Democrat",
    Q30_combo_45na == 3 ~ "Independent",
    Q30_combo_45na == 4 ~ "Other/NA",
    TRUE ~ NA_character_
  ))
```

```r
full_model <- glm(
  infrequent_voter ~ Q30_combo_45na + ppage + educ + race + gender +
 ↪  income_cat,
  data = voter_data,
  family = "binomial"
)

null_model <- glm(
  infrequent_voter ~ 1,
  data = voter_data,
  family = "binomial"
)

stepwise_model <- step(
  null_model,
  scope = list(lower = null_model, upper = full_model),
  direction = "both"
)
```

```
Start:  AIC=5954.42
infrequent_voter ~ 1

                 Df Deviance    AIC
+ ppage           1   5365.6 5369.6
+ Q30_combo_45na  3   5679.6 5687.6
+ income_cat      3   5797.8 5805.8
+ educ            2   5806.6 5812.6
+ race            3   5931.1 5939.1
+ gender          1   5949.6 5953.6
<none>                5952.4 5954.4

Step:  AIC=5369.63
```

```
infrequent_voter ~ ppage

                 Df Deviance    AIC
+ educ            2   5107.5 5115.5
+ Q30_combo_45na  3   5175.0 5185.0
+ income_cat      3   5189.2 5199.2
<none>                5365.6 5369.6
+ race            3   5360.5 5370.5
+ gender          1   5365.3 5371.3
- ppage           1   5952.4 5954.4

Step:  AIC=5115.45
infrequent_voter ~ ppage + educ

                 Df Deviance    AIC
+ Q30_combo_45na  3   4973.7 4987.7
+ income_cat      3   5046.5 5060.5
+ race            3   5099.7 5113.7
+ gender          1   5104.9 5114.9
<none>                5107.5 5115.5
- educ            2   5365.6 5369.6
- ppage           1   5806.6 5812.6

Step:  AIC=4987.7
infrequent_voter ~ ppage + educ + Q30_combo_45na

                 Df Deviance    AIC
+ income_cat      3   4926.2 4946.2
+ race            3   4966.7 4986.7
+ gender          1   4971.2 4987.2
<none>                4973.7 4987.7
- Q30_combo_45na  3   5107.5 5115.5
- educ            2   5175.0 5185.0
- ppage           1   5573.3 5585.3

Step:  AIC=4946.16
infrequent_voter ~ ppage + educ + Q30_combo_45na + income_cat

                 Df Deviance    AIC
+ gender          1   4922.6 4944.6
+ race            3   4919.0 4945.0
<none>                4926.2 4946.2
- income_cat      3   4973.7 4987.7
- Q30_combo_45na  3   5046.5 5060.5
- educ            2   5045.1 5061.1
- ppage           1   5524.5 5542.5

Step:  AIC=4944.55
```

```
infrequent_voter ~ ppage + educ + Q30_combo_45na + income_cat +
    gender

                 Df Deviance    AIC
+ race            3   4915.3 4943.3
<none>                4922.6 4944.6
- gender          1   4926.2 4946.2
- income_cat      3   4971.2 4987.2
- Q30_combo_45na  3   5042.4 5058.4
- educ            2   5042.5 5060.5
- ppage           1   5513.6 5533.6

Step:  AIC=4943.26
infrequent_voter ~ ppage + educ + Q30_combo_45na + income_cat +
    gender + race

                 Df Deviance    AIC
<none>                4915.3 4943.3
- race            3   4922.6 4944.6
- gender          1   4919.0 4945.0
- income_cat      3   4964.1 4986.1
- Q30_combo_45na  3   5033.9 5055.9
- educ            2   5037.3 5061.3
- ppage           1   5495.8 5521.8
```

```r
summary(stepwise_model)
```

```
Call:
glm(formula = infrequent_voter ~ ppage + educ + Q30_combo_45na +
    income_cat + gender + race, family = "binomial", data = voter_data)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)               0.470446   0.193551   2.431  0.01507 *
ppage                    -0.052067   0.002339 -22.265  < 2e-16 ***
educHigh school or less   1.015572   0.097039  10.466  < 2e-16 ***
educSome college          0.276589   0.095107   2.908  0.00364 **
Q30_combo_45na2          -0.112760   0.102270  -1.103  0.27021
Q30_combo_45na3           0.280606   0.102439   2.739  0.00616 **
Q30_combo_45na4           0.962308   0.110915   8.676  < 2e-16 ***
income_cat$40-75k         0.060157   0.111739   0.538  0.59032
income_cat$75-125k       -0.061771   0.106832  -0.578  0.56313
income_catLess than $40k  0.583500   0.113390   5.146 2.66e-07 ***
genderMale                0.139378   0.072197   1.931  0.05354 .
raceHispanic              0.088271   0.129205   0.683  0.49449
raceOther/Mixed           0.397583   0.161667   2.459  0.01392 *
```

```
raceWhite                 0.035100   0.108528   0.323  0.74638
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5952.4  on 5335  degrees of freedom
Residual deviance: 4915.3  on 5322  degrees of freedom
AIC: 4943.3

Number of Fisher Scoring iterations: 5
```